

ModSwap Design Walkthrough

Mobile (iOS / Android) • Web • Flutter Material 3

Prepared by the ModSwap Engineering Team • 31 May 2026 • Private

Complete screen-by-screen walkthrough of the ModSwap student marketplace application across mobile and web platforms. Covers authentication (including biometric login), home & semantic search, item listing and management, wishlist, real-time notifications, and profile & settings. Each screen documents its purpose, key UI elements, and user flow.

ModSwap Engineering Team

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KMUTT • Faculty of Information
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Academic Year 2025–2026

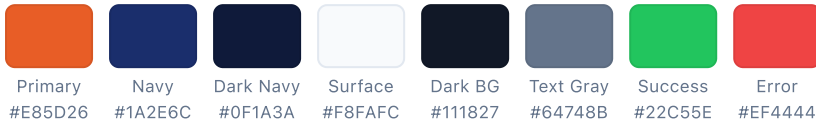
Contents

1	Design Language	2
	Color Palette • Typography • Component Library	
2	Authentication Flow	3
	Login • Register • Complete Profile • Forgot Password • Biometric Login	
3	Home & Discovery	4
	Home Screen • Semantic Search • Filter by Type • Dark Mode	
4	Item Detail	5
	Listing Detail View • Add to Wishlist	
5	Post & Edit Item	6
	Post New Item • Category Picker • Meeting Points • Edit Item	
6	My Items	7
	Drafts • Published • Sold • Mark as Sold	
7	Wishlist	8
8	Notifications	9
	Login Alert • Wishlist • Price Drop • Deal Complete • Rating • Rate Prompt	
9	Profile & Settings	10
	Menu/Profile • Edit Profile • Change Password • Dark Mode • Logout	
10	Web Responsive Layout	11
	Home • Login • Register • My Items • Profile • Notifications • Dark Mode	

1 Design Language

ModSwap follows Material 3 guidelines with a custom brand identity. The palette, typography, and component system are consistent across mobile and web platforms.

1.1 Color Palette



1.2 Typography

Aa

Display 24pt Bold
Screen titles

Aa

Heading 16pt SemiBold
Section headers

Aa

Subheading 13pt Medium
Card titles, labels

Aa

Body 13–14pt Regular
Descriptions, hints

Aa

Caption 11–12pt Regular
Timestamps, metadata

1.3 Core UI Components

Buttons

- Primary: Orange filled, pill shape (radius 25)
- Secondary: Navy outlined, pill shape
- Destructive: Red text, no background
- Loading: CircularProgressIndicator inline

Input Fields

- Rounded pill border (radius 28)
- Gray border at rest, orange on focus
- Red border on validation error
- Prefix icons in gray, suffix eye toggle

Navigation

- Bottom nav: Home / My Item / + / Notification / Menu
- Center FAB (+) for post new item
- Badge counter on Notification icon
- Active tab highlighted in orange

Cards & Grids

- Item cards: image + title + price + category chip
- 2-column grid on home, white bg + subtle shadow
- SWAP badge on price when open to swap
- DRAFT / SOLD overlays on My Items cards

Bottom Sheets

- Modal sheets for pickers and quick actions
- Drag handle at top center
- Category, Meeting Points, Filter, Add Photo
- Rounded top corners (radius 20)

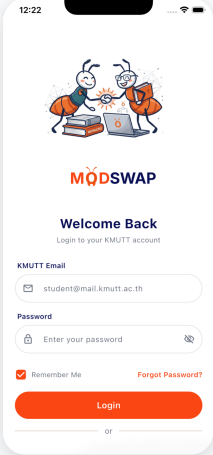
Notifications

- Filter chips: All / Buyer / For You / System
- Orange active chip, gray inactive
- Color-coded icons per notification type
- Orange dot for unread indicator

2 Authentication Flow

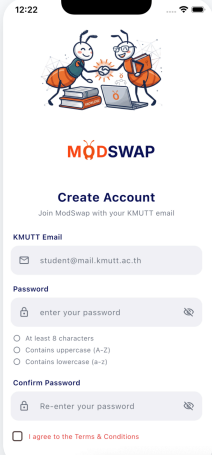
All accounts are restricted to @mail.kmutt.ac.th emails. The flow covers registration, login, password recovery, and biometric quick-login via local_auth + flutter_secure_storage.

2.1 Login & Register



The Login Screen features the MōdSwap mascot logo at the top. Below it, the text 'Welcome Back' and 'Login to your KMUTT account' are displayed. There are two input fields: 'KMUTT Email' with the placeholder 'student@mail.kmutt.ac.th' and 'Password' with the placeholder 'Enter your password'. A 'Remember Me' checkbox is checked, and a 'Forgot Password?' link is visible. A large orange 'Login' button is at the bottom.

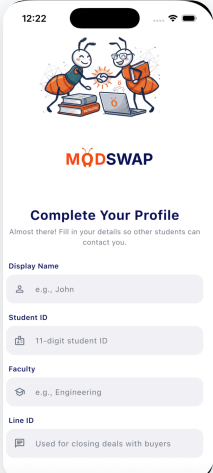
Purpose	Authenticate an existing KMUTT student account
Key Elements	Mascot logo • KMUTT Email field • Password with show/hide toggle • Remember Me checkbox • Forgot Password link • Login button • Register link
User Flow	Enter @mail.kmutt.ac.th email → password → Login → Home. Inline error shown below email on failure.



The Register Screen shows the MōdSwap mascot logo and the text 'Create Account' and 'Join ModSwap with your KMUTT email'. It includes an email field with 'student@mail.kmutt.ac.th', a password field with 'enter your password', and a 'Confirm Password' field with 'Re-enter your password'. There are checkboxes for password requirements: 'At least 8 characters', 'Contains uppercase (A-Z)', and 'Contains lowercase (a-z)'. A checkbox for 'I agree to the Terms & Conditions' is at the bottom.

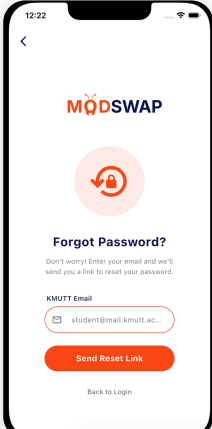
Purpose	Create a new ModSwap account with a KMUTT email
Key Elements	Email field • Password with live validation indicators (8+ chars, uppercase, lowercase) • Confirm Password • Terms & Conditions checkbox
User Flow	Fill email + password → Check T&C → Register → Verification email sent → Verify Email screen

2.2 Complete Profile & Forgot Password



The 'Complete Your Profile' screen has the MōdSwap mascot logo and the text 'Complete Your Profile' and 'Almost there! Fill in your details so other students can contact you.' It contains four input fields: 'Display Name' (placeholder 'e.g., John'), 'Student ID' (placeholder '11-digit student ID'), 'Faculty' (placeholder 'e.g., Engineering'), and 'Line ID' (placeholder 'Used for closing deals with buyers').

Purpose	Collect identity details required for campus transactions
Key Elements	Display Name • Student ID (11-digit) • Faculty • Line ID (used to close deals)
User Flow	Shown once after email verification → Fill all fields → Save → Firestore write → Home

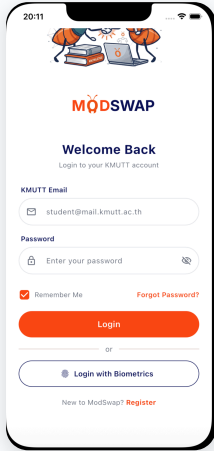
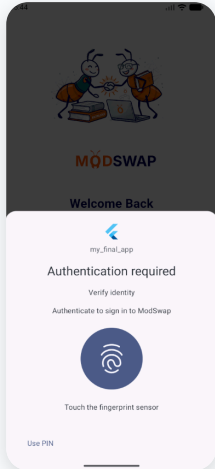
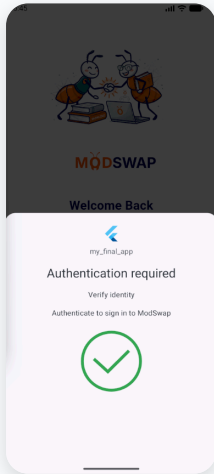
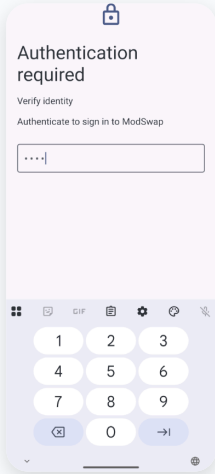


The 'Forgot Password' screen features the MōdSwap mascot logo and the text 'Forgot Password?' and 'Don't worry! Enter your email and we'll send you a link to reset your password.' It includes an email field with 'student@mail.kmutt.ac...' and a large orange 'Send Reset Link' button. A 'Back to Login' link is at the bottom.

Purpose	Trigger Firebase password-reset email
Key Elements	Lock icon illustration • KMUTT Email field (constrained to @mail.kmutt.ac.th) • Send Reset Link button • Back to Login link
User Flow	Enter registered email → Send Reset Link → Success toast → User checks inbox

2.3 Biometric Login Flow

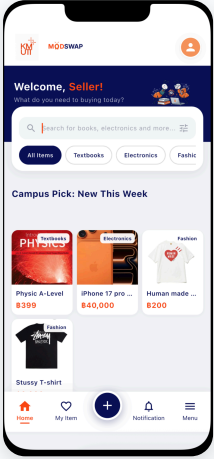
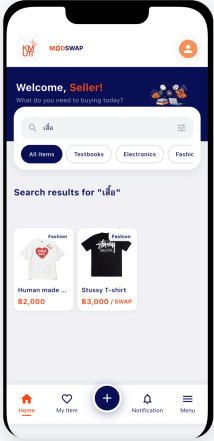
After first login with "Remember Me", credentials are saved encrypted via flutter_secure_storage. On next launch the biometric button appears and triggers the OS authentication dialog.

	① Login with Biometrics <table><tr><td>Purpose</td><td>Allow returning users to log in without re-entering email and password</td></tr><tr><td>Key Elements</td><td>"Login with Biometrics" outlined button (navy border, fingerprint icon) • Visible only when device supports biometrics AND "Remember Me" was checked on first login</td></tr><tr><td>User Flow</td><td>Tap "Login with Biometrics" → OS authentication dialog opens → On success, saved credentials are loaded and login proceeds automatically</td></tr></table>	Purpose	Allow returning users to log in without re-entering email and password	Key Elements	"Login with Biometrics" outlined button (navy border, fingerprint icon) • Visible only when device supports biometrics AND "Remember Me" was checked on first login	User Flow	Tap "Login with Biometrics" → OS authentication dialog opens → On success, saved credentials are loaded and login proceeds automatically
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	② Fingerprint Prompt <table><tr><td>Purpose</td><td>Verify user identity via device biometric hardware</td></tr><tr><td>Key Elements</td><td>OS-native system dialog • App name (my_final_app) • "Authenticate to sign in to ModSwap" message • Fingerprint sensor icon • "Use PIN" fallback link at bottom</td></tr><tr><td>User Flow</td><td>Touch fingerprint sensor → Sensor reads fingerprint → Match triggers success state • Sensor failure prompts retry or PIN fallback</td></tr></table>	Purpose	Verify user identity via device biometric hardware	Key Elements	OS-native system dialog • App name (my_final_app) • "Authenticate to sign in to ModSwap" message • Fingerprint sensor icon • "Use PIN" fallback link at bottom	User Flow	Touch fingerprint sensor → Sensor reads fingerprint → Match triggers success state • Sensor failure prompts retry or PIN fallback
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	③ Authentication Success <table><tr><td>Purpose</td><td>Confirm successful biometric verification to the user</td></tr><tr><td>Key Elements</td><td>Green circle checkmark animation • Same OS dialog backdrop • No user action required — app proceeds automatically</td></tr><tr><td>User Flow</td><td>Fingerprint matched → Green checkmark shown briefly → Dialog dismisses → App calls login with saved credentials → Navigate to Home screen</td></tr></table>	Purpose	Confirm successful biometric verification to the user	Key Elements	Green circle checkmark animation • Same OS dialog backdrop • No user action required — app proceeds automatically	User Flow	Fingerprint matched → Green checkmark shown briefly → Dialog dismisses → App calls login with saved credentials → Navigate to Home screen
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	④ PIN Fallback <table><tr><td>Purpose</td><td>Allow authentication when fingerprint fails or is unavailable</td></tr><tr><td>Key Elements</td><td>OS PIN entry screen • Lock icon at top • "Authentication required" heading • PIN dot input field • Full numeric keypad</td></tr><tr><td>User Flow</td><td>Tap "Use PIN" on fingerprint dialog → PIN screen shown • Enter device PIN → On match, same success flow as fingerprint — credentials loaded, login proceeds</td></tr></table>	Purpose	Allow authentication when fingerprint fails or is unavailable	Key Elements	OS PIN entry screen • Lock icon at top • "Authentication required" heading • PIN dot input field • Full numeric keypad	User Flow	Tap "Use PIN" on fingerprint dialog → PIN screen shown • Enter device PIN → On match, same success flow as fingerprint — credentials loaded, login proceeds
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User Flow	Tap "Use PIN" on fingerprint dialog → PIN screen shown • Enter device PIN → On match, same success flow as fingerprint — credentials loaded, login proceeds						

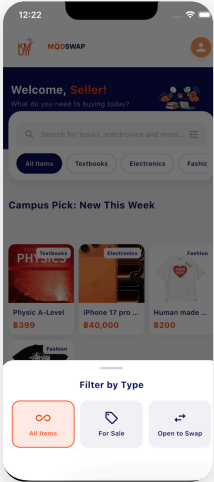
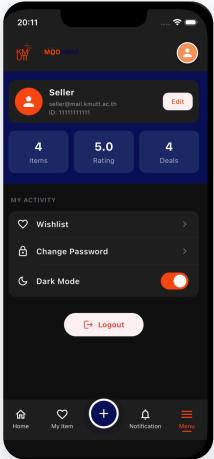
3 Home & Discovery

Primary browsing entry point. Features Gemini-powered semantic search (controlled by Remote Config flag `enable_semantic_search`), category quick-filter chips, and a "Campus Pick: New This Week" grid.

3.1 Home Screen

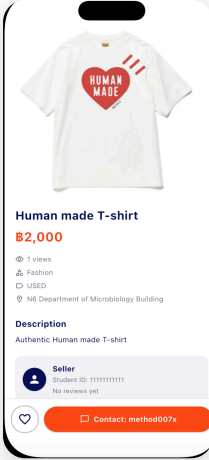
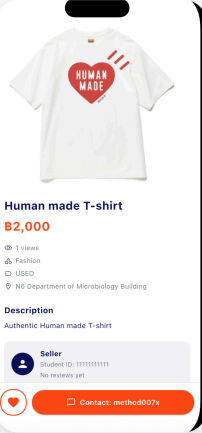
	Home — Light Mode	
	Purpose	Browse all marketplace listings and discover new campus items
	Key Elements	KMUTT logo + ModSwap wordmark • Personalized greeting • Search bar with filter icon • Category chips (All Items, Textbooks, Electronics, Fashion, Housing/Dorm, Vehicles) • 2-column item grid • Bottom nav
	Semantic Search Results	
	Purpose	Natural language search — Gemini embedding-001 matches semantic intent, not just keywords
	Key Elements	Active search bar with query • "Search results for [query]" label • Matching items grid • Thai language supported (e.g. "เสื้อ" finds T-shirts)
User Flow		Land after login → Browse grid → Tap category chip to filter → Tap item card → Item Detail
User Flow		Tap search bar → Type Thai or English query → Results load → Tap item

3.2 Filter & Dark Mode

	Filter by Type	
	Purpose	Narrow browse results by listing type
	Key Elements	Modal bottom sheet • Three tiles: All Items / For Sale / Open to Swap • Selected tile has orange border
User Flow		Tap filter icon (≡) in search bar → Sheet slides up → Tap tile → Grid filters instantly
	Dark Mode — Home	
	Purpose	Reduce eye strain in low-light; stored as user preference
	Key Elements	Dark surface (#111827) replaces white backgrounds • All text, icons, and accent colors preserved • Toggle in Menu screen
User Flow		Menu → Toggle "Dark Mode" → Entire app switches immediately, persisted across restarts

4 Item Detail

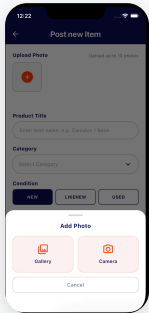
Full listing view with all details needed to evaluate and contact a seller. Wishlist heart button writes to Firestore in real-time and triggers a seller notification.

	<table><tr><td>Purpose</td><td>Full information view — lets buyer evaluate and contact seller</td></tr><tr><td>Key Elements</td><td>Hero product image (scrollable) • Title + Price (orange) • View count • Category • Condition badge • KMUTT meeting location • Description • Seller card (name, Student ID, rating) • Contact button (opens Line) • Heart wishlist button</td></tr><tr><td>User Flow</td><td>Tap item card → View details → Contact to chat on Line • Tap heart → Saved to Wishlist; seller notified</td></tr></table>	Purpose	Full information view — lets buyer evaluate and contact seller	Key Elements	Hero product image (scrollable) • Title + Price (orange) • View count • Category • Condition badge • KMUTT meeting location • Description • Seller card (name, Student ID, rating) • Contact button (opens Line) • Heart wishlist button	User Flow	Tap item card → View details → Contact to chat on Line • Tap heart → Saved to Wishlist; seller notified
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	<table><tr><td>Purpose</td><td>Confirm wishlist state; allow buyer to unsave</td></tr><tr><td>Key Elements</td><td>Heart icon filled red (vs. outlined default) • Real-time Firestore write on tap</td></tr><tr><td>User Flow</td><td>Tap outlined heart → Turns red → Seller gets "Someone Wishlisted Your Item" notification • Tap red heart to remove</td></tr></table>	Purpose	Confirm wishlist state; allow buyer to unsave	Key Elements	Heart icon filled red (vs. outlined default) • Real-time Firestore write on tap	User Flow	Tap outlined heart → Turns red → Seller gets "Someone Wishlisted Your Item" notification • Tap red heart to remove
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5 Post & Edit Item

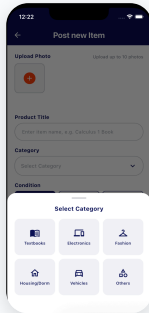
Sellers create listings via the center FAB (+). The form supports up to 10 photos, 6 categories, 3 condition states, optional swap toggle, and KMUTT-specific meeting points. Items are saved as Draft before being Published.

5.1 Post New Item — Pickers



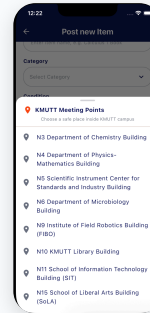
Add Photo Sheet

Purpose	Choose image source for listing photos
Elements	Photo grid (up to 10) • Bottom sheet: Gallery / Camera • Cancel
Flow	Tap + → Sheet appears → Choose source → Photo added



Category Picker

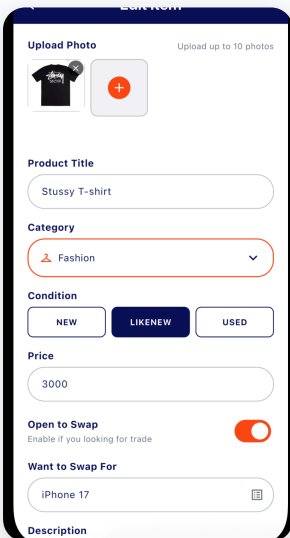
Purpose	Tag listing category for search and filter
Elements	6 tiles: Textbooks, Electronics, Fashion, Housing/Dorm, Vehicles, Others
Flow	Tap Category field → Sheet slides up → Tap tile → Category set



Meeting Points Picker

Purpose	Set a safe KMUTT campus building as deal location
Elements	Buildings N3–N15 with pin icons • "Choose a safe place inside KMUTT campus"
Flow	Tap Meeting Location → Sheet shows buildings → Tap row to select

5.2 Edit Item



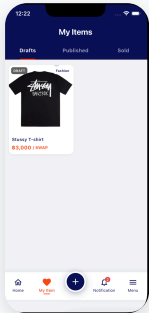
Edit Item Screen

Purpose	Modify an existing draft or published listing
Key Elements	Photo thumbnail with remove (x) • Product Title field • Category dropdown (active shows orange border) • Condition toggle (NEW / LIKENEW / USED) • Price field • Open to Swap toggle • Want to Swap For field • Description text area
User Flow	My Items → Tap draft card → Modify fields → Save as Draft or Publish. Updates reflected via Firestore real-time.

6 My Items

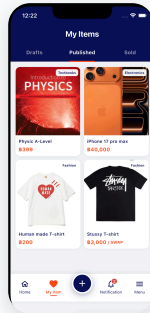
Sellers manage inventory through three tabs: Drafts (saved, not live), Published (active listings), and Sold (completed deals). Mark as Sold records deal type, final price, and buyer identity for rating generation.

6.1 Item States



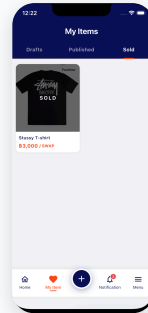
Drafts Tab

Purpose	Items saved but not yet published
Elements	DRAFT badge on card • Category chip • Title + price + SWAP tag
Flow	Tap card → Edit Item → Publish when ready



Published Tab

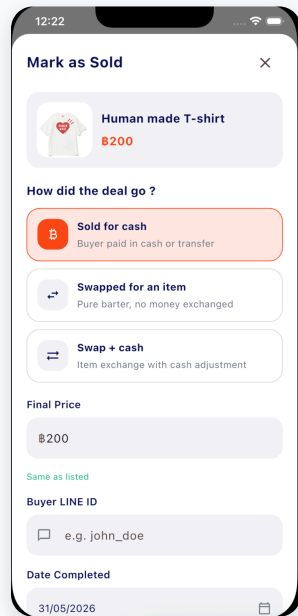
Purpose	All active listings visible to other students
Elements	Category badges • Price with SWAP indicator • Edit / Mark-as-Sold actions
Flow	Tap item → Edit or Mark as Sold; live on Firestore



Sold Tab

Purpose	Archive of completed transactions
Elements	SOLD overlay on grayed card • Original price shown • Read-only state
Flow	Moved here automatically after Mark as Sold is confirmed

6.2 Mark as Sold

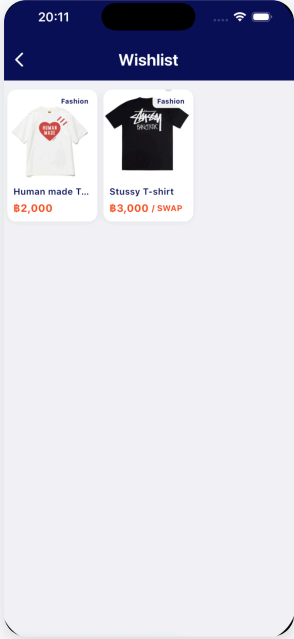


Mark as Sold Modal

Purpose	Record how the deal was completed and who the buyer was, enabling rating prompt generation
Key Elements	Item preview card • Deal type: Sold for cash / Swapped for an item / Swap + cash • Final Price field (pre-filled from listing) • Buyer LINE ID field • Date Completed picker • Confirm button
User Flow	Tap item → Mark as Sold → Select deal type → Confirm → Item moves to Sold tab • Buyer gets "Deal Complete" notification • Rating prompt appears on buyer's next home launch

7 Wishlist

Students save items of interest. Price drop notifications fire automatically when a wishlisted item's price is reduced by the seller.



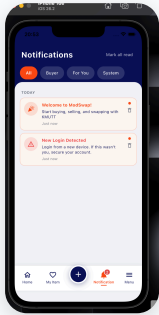
Wishlist Screen

Purpose	Central view of all saved listings for quick access and price monitoring
Key Elements	Navy app bar • 2-column item grid • Category badge per card • Title + price • SWAP badge if open to swap • Tap card → Item Detail
User Flow	Menu → Wishlist • Or: Item Detail → Heart button → Saved here • Price drop → Notification badge appears

8 Notifications

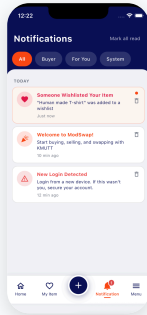
Real-time inbox powered by Firestore streams via Riverpod. Four filter tabs: **All**, **Buyer**, **For You** (wishlist/price alerts), **System** (security). "Mark all read" clears the bottom nav badge.

8.1 Notification Types



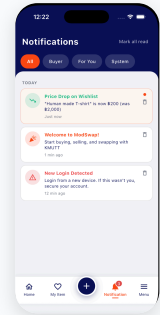
Login / System Alert

Type	System
Trigger	New login from an unrecognized device
Message	"New Login Detected — If this wasn't you, secure your account." + Welcome notification



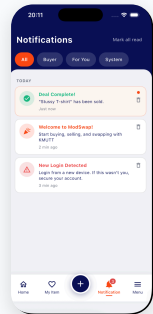
Wishlist Alert

Type	For You (seller)
Trigger	Another user wishlists your item
Message	"Someone Wishlisted Your Item — [item] was added to a wishlist"



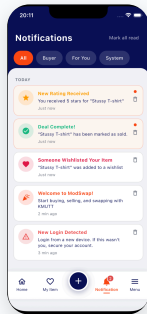
Price Drop Alert

Type	For You (buyer)
Trigger	Seller reduces price of a wishlisted item
Message	"Price Drop on Wishlist — [item] is now \$X (was \$Y)"



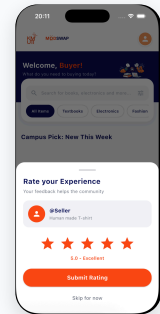
Deal Complete

Type	Buyer
Trigger	Seller marks item as sold to this buyer
Message	"Deal Complete! — [item] has been sold."



Rating Received

Type	For You (seller)
Trigger	Buyer submits a star rating after a deal
Message	"New Rating Received — You received 5 stars for [item]"



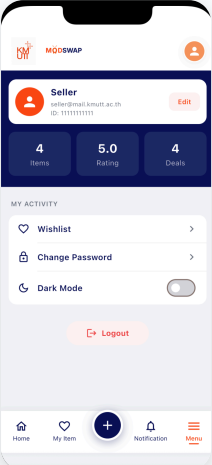
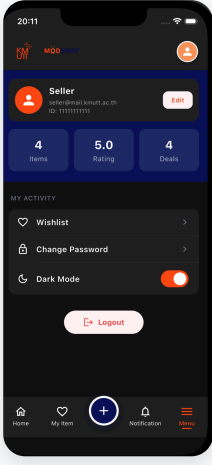
Rate your Experience

Type	Buyer prompt (modal overlay)
Trigger	Buyer opens app after deal is completed
Elements	Seller avatar • 5-star selector • Submit Rating • Skip for now

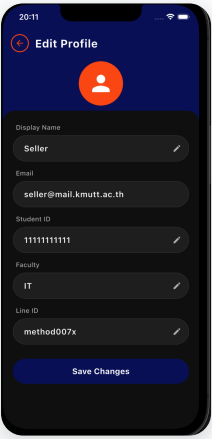
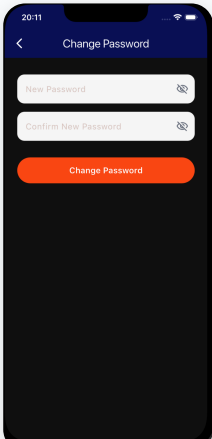
9 Profile & Settings

The Menu tab is the profile hub showing user stats, settings access, and dark mode toggle. Edit Profile allows updating display name, faculty, and Line ID. Email is locked (KMUTT account cannot be changed).

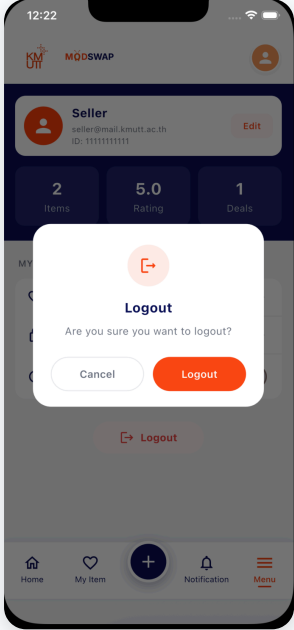
9.1 Menu & Dark Mode

	Menu — Light Mode	
	Purpose	Central hub for user identity, stats, and settings
	Key Elements	Profile card: avatar, name, email, Student ID, Edit button • Stats grid: Items / Rating / Deals • My Activity: Wishlist, Change Password, Dark Mode toggle • Logout button
	User Flow	Tap Menu tab → Edit profile • Wishlist for saved items • Toggle Dark Mode • Logout with confirmation dialog
	Menu — Dark Mode Active	
	Purpose	Same menu in dark theme — orange toggle indicates active state
	Key Elements	Dark surface backgrounds • Text and icon contrasts maintained • Stats cards in dark navy • Toggle ON (orange)
	User Flow	Toggle switches instantly → All screens update → Preference persisted across restarts

9.2 Edit Profile & Change Password

	Edit Profile	
	Purpose	Update personal info shown to other users and used in deals
	Key Elements	Avatar (tappable) • Display Name (editable) • Email (read-only) • Student ID • Faculty • Line ID • Save Changes button
	User Flow	Tap Edit on profile card → Modify fields → Save Changes → Written to Firestore. Email field is locked.
	Change Password	
	Purpose	Update account password from within the app
	Key Elements	New Password field with show/hide • Confirm New Password • Change Password button (orange) • Dark mode active
	User Flow	Menu → Change Password → Enter + confirm new password → Firebase Auth credential updated → Back to Menu

9.3 Logout Confirmation



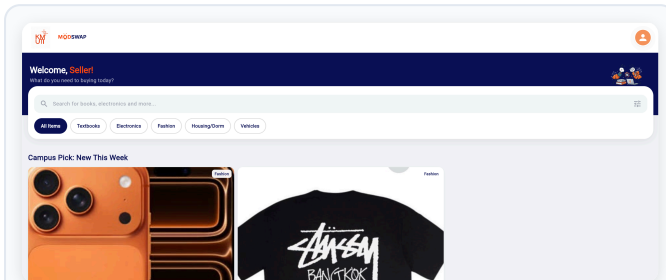
Logout Dialog

Purpose	Confirm intent before signing out to prevent accidental logouts
Key Elements	Centered modal dialog • Logout icon in orange circle • "Are you sure you want to logout?" • Cancel (outlined) + Logout (orange filled) buttons
User Flow	Tap Logout → Dialog appears → Cancel to dismiss • Confirm → Firebase session cleared → Login screen

10 Web Responsive Layout

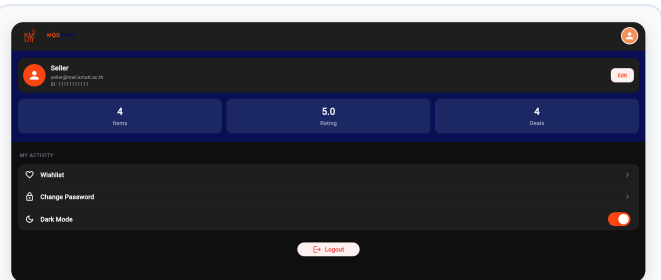
ModSwap runs as a Flutter Web app sharing the same codebase as mobile. Layout adapts to desktop viewport with fluid grid, full-width nav bar, and expanded content areas. All features including dark mode are fully supported.

10.1 Home (Web)



Home — Web Light Mode

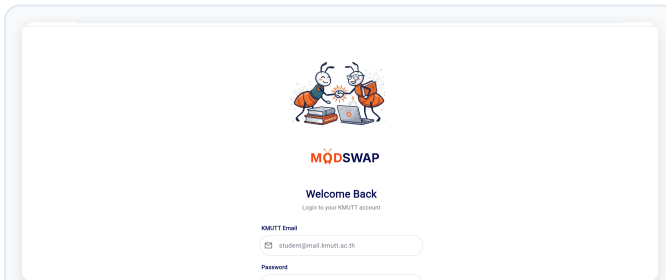
Purpose	Full desktop home with wider grid layout
Key Elements	Full-width top bar • Welcome banner with mascot • Full-width search bar • Category chips in row • Multi-column item grid • Bottom nav same as mobile



Home — Web Dark Mode

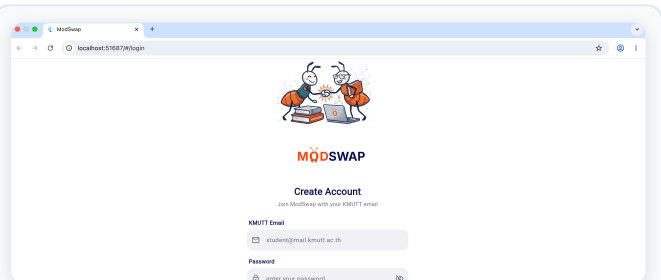
Purpose	Dark theme on desktop — identical toggle behavior to mobile
Key Elements	Dark background across full viewport • Item cards use dark surface • All contrasts preserved • Orange accent unchanged

10.2 Login & Register (Web)



Login — Web

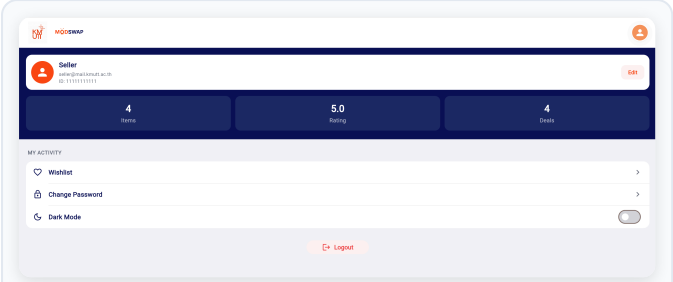
Purpose	Browser-based login — centered on white canvas
Key Elements	Mascot logo centered • Max-width 420px card • Identical fields to mobile: Email, Password, Remember Me, Forgot Password • Login button • Register link
Note	Biometric button not shown on web (device API unavailable in browser)



Register — Web

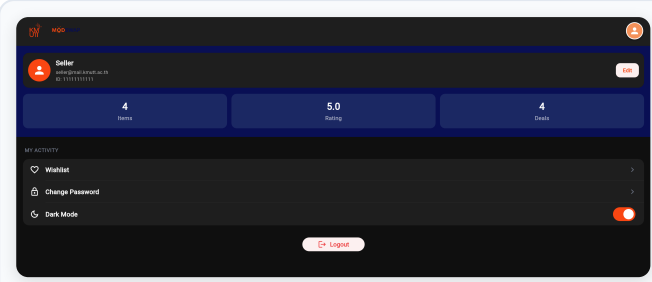
Purpose	Account creation in browser — identical validation to mobile
Key Elements	Create Account heading • Email, Password (with live validation), Confirm Password • T&C checkbox • Register button

10.3 Profile, My Items & Notifications (Web)



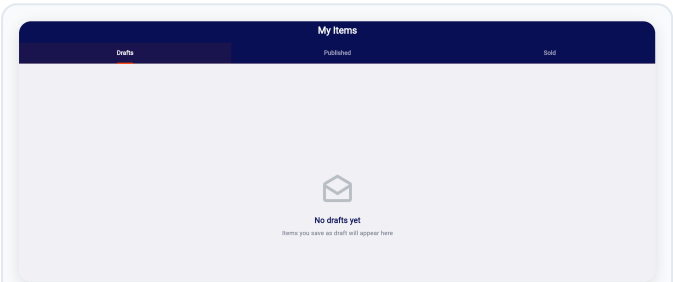
My Profile — Web Light

Purpose	Full-width profile view on desktop
Key Elements	Profile card spanning full width • Three-column stats bar (Items / Rating / Deals) • My Activity list • Dark Mode toggle • Logout centered



My Profile — Web Dark

Purpose	Profile in dark mode — consistent theming on web
Key Elements	Dark surface for all cards • Stats grid dark navy • Orange toggle active • Text contrast maintained



My Items — Web

Purpose	Seller item management on desktop
Key Elements	Navy header full-width • Drafts / Published / Sold tabs • Empty state: envelope icon + "No drafts yet" • FAB (+) centered in bottom bar



Notifications — Web

Purpose	Full notification inbox on desktop
Key Elements	Filter chips (All / Buyer / For You / System) • Full-width notification rows • All types visible: Login, Rating, Deal, Wishlist, Price Drop • Trash icon per row • Mark all read